

LIAM NANCE

Graduate Aerospace Engineer | Propulsion | Compressible Flow | Astronautics | **U.S Citizen**

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EDUCATION

Arizona State University

Tempe, AZ

MS - Aerospace Engineering

Expected May 2026

- GPA: 4.00/4.00
- Focused Topics: Rocket & Aircraft Propulsion, CFD, Aircraft Structures, Spacecraft Systems, FEM, Aerodynamics, Controls

BSE - Aerospace Engineering

Graduated May 2025

- GPA: 3.91/4.00, *Summa Cum Laude*

SKILLS

Engineering Software: MATLAB, Python, Siemens NX, SolidWorks, Fusion 360, Ansys Fluent (CFD), XFOIL, Arduino, AutoCAD

Technical Domains: GD&T (ASME Y14.5-2009), CAD, CFD, Aerodynamics, Propulsion, Compressible Flow, Thermodynamics

Manufacturing: CNC Router, CNC Mill, Laser Cutting, Toolpathing

PROGRAMS AND PROJECTS

Aphrodite 1 - Theoretical Rotating Space Station

Aug, 2024 - May, 2025

- Thermal Analysis lead. Designed a collapsible radiator system integrated with internal fluid loops and heat exchangers.
- Designed an 8-DOF robotic arm for in-orbit assembly and maintenance using CAD and 3D printing.
- Performed materials and coatings trade studies based on thermal emissivity, longevity, and MMOD resistance.

Thorough Analysis of Ramjet and Scramjet Flow - MATLAB

Dec, 2025

- Developed a MATLAB-based performance tool for non-isentropic ram-compression Brayton Cycle engines.
- Enabled parametric studies on Mach number, flight altitude, internal flow speeds, maximum material temperatures, isentropic efficiencies, pressures, ISP, and TSFC for engine optimization.

Spaceworks at ASU - NASA JPL Partner Program

Aug, 2024 - Aug, 2025

- Applied the NASA standard procedure and documents in a professional team setting. Visited the JPL Campus.
- Lead engineer and designer for the construction of an "Earth Lander" which dropped from 100+ feet, and successfully delivered a payload to the ground. Used CAD and CFD for systems design, in addition to physical modeling.

Rigorous Analysis of Liquid Rocket Engine (LRE)

Dec, 2024

- Calculated flow properties of LOX/LH2 at 15+engine locations, including preburners, turbines, injectors, and nozzle.
- Identified key engine parameters such as thrust, temperature limit, maximum flow velocity, and engine efficiency.
- Analyzed LRE performance trends, limits, and nuanced internal interactions.

3D Cut CNC Tambour Sliding Door Cabinet

- Designed and modeled cabinet/tambour system in Fusion 360; created toolpaths in VCarve, and CNC-machined components using a CNC router.
- Iterated design, tolerances, and channel geometry to optimize form, function, and reliability.

Boy Scouts of America - Eagle Scout

- Eagle Scout with 10+ years of leadership, project planning, and team coordination experience.
- Completed 500+ hours of community service, including planning and executing large-scale service projects.

WORK EXPERIENCE

Chas Roberts Plumbing and AC

Peoria, AZ

CAD Drafter

Aug, 2024 - Present

- Design detailed AutoCAD HVAC drawings containing hundreds of components, adhering to engineering specifications.
- Review technical drawings and specification sheets to ensure proper system integration from design through implementation.

Greystar Real Estate Partners

Tempe, AZ

Leasing Assistant

Mar, 2023 - Jun, 2024

- Learned to communicate in a team environment and facilitated over \$500,000 in leases for the property.

Microsoft

Tempe, AZ

Support Technician - SMB

Feb, 2022 - Mar, 2023

- Diagnosed and resolved complex system and software issues using root cause analysis and internal research—managed high-stakes support cases, mitigating time and financial loss.